Intelligent surgical workflow recognition-based skill assessment for minimally invasive submandibular gland resection.

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Abstract

Artificial Intelligence (AI) is transforming clinical dental practice, with most applications focusing on static images like radiographs. However, the integration of AI in analyzing video data, particularly in surgical settings rich in dynamic information, remains underexplored in oral medicine. This study introduces AI-miSMG, an intelligent surgical workflow recognition model designed for minimally invasive submandibular gland (SMG) resection. Building upon our previous work, we established a standardized annotation protocol to efficiently label endoscopic SMG resection procedures, which are divided into Creation, Position, Separation, Inspection, and Idle phases. The AI-miSMG model was trained on a dataset of 73 high-quality endoscopic SMG resection videos, comprising 386,122 labeled frames. To evaluate its performance, we used a multicenter dataset consisting of surgical videos from four different centers, totaling 85,913 images. The model achieved an overall accuracy of 0.87 on the external validation dataset. Additionally, it was explored for workflow-based analysis of surgical fluency across surgeons with different experience levels. Furthermore, our model reduced annotation time by approximately 47%, decreasing from 94.00 min to 49.90 min. Overall, we developed a deep learning-based workflow recognition model for minimally invasive SMG resection and demonstrated its feasibility for surgical phase recognition, workflow analysis, and model-assisted annotation.

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